6 Discussion

How much of measured connectedness is a common factor? A single factor explains on average 58% of the variance of weekly returns, and removing it lowers total connectedness by about 41% (38–43% across all robustness checks). This number, however, understates the role of the factor. Diebold–Yilmaz connectedness is a share of the forecast-error variance of the returns, whereas FNETS connectedness is a share of the forecast-error variance of the idiosyncratic component only, which is itself about 42% of the total. As a rough calculation, idiosyncratic spillovers account for about $0.47 \times 0.42 \approx 20\%$ of the variance of a typical market, against the 81% that Diebold–Yilmaz attributes to shocks in other markets. Most of what the dense measure reports as connectedness is therefore exposure to the same shock rather than transmission between markets. This is consistent with a global financial cycle (Rey, 2013): the factor is not simply the US market, since excluding the S&P 500 leaves both the common share and the loss of connectedness unchanged, so it is a factor common to emerging markets as a group. It is also consistent with Forbes and Rigobon (2002). The peak of Diebold–Yilmaz connectedness in the windows containing March 2020 coincides with a peak of the common share, while idiosyncratic connectedness shows no peak at all: what looks like contagion in the dense measure is an increase in the weight of the common shock, together with higher volatility, and not an increase in bilateral links. The same reading applies to the decline after 2020, which comes with a smaller common share (58% to 47%) and a slightly higher idiosyncratic connectedness. The high correlation (0.95) between the common share and Diebold–Yilmaz connectedness is partly mechanical, because both increase whenever all pairwise correlations increase; it shows that the two quantities summarize the same feature of the data, not that one causes the other.

Does the method change the picture? Regularization alone does not. The lasso sets almost all lag coefficients to zero and still reproduces Diebold–Yilmaz connectedness (correlation 0.98), and the horizon has virtually no effect. Both facts have the same explanation: at the weekly frequency the moving-average coefficients Ψ_h for $h \geq 1$ are close to zero, so the generalized variance shares in equation (1) reduce to approximately ρ_{ij}^2, the squared correlations, normalized by row. In this setting Diebold–Yilmaz connectedness is close to a normalized squared-correlation matrix, the kind of pairwise and contemporaneous measure that Diebold and Yılmaz (2014) set out to improve on. The approximating model adds little dynamic content here because the data contain little. NETS keeps the same time pattern at a lower level because it regularizes the innovation concentration matrix, not only the lags, and sets weak partial correlations to zero; the connectedness it removes is mostly diffuse, small links. FNETS measures a different object, and its low agreement with the other methods is expected by construction. The choice of method should therefore follow the purpose. For monitoring how tightly markets move together, Diebold–Yilmaz or the lasso VAR are preferable: they are smooth, robust to every specification we tried, and simple to compute. For portfolio diversification across emerging markets, the relevant quantity is the common-factor share, because that part of the co-movement cannot be diversified away within this universe. For the study of contagion, only a factor-adjusted measure such as FNETS addresses the right question, but in our setting it is also the least reliable estimate.

Transmitters and receivers. South Africa, Korea and Taiwan are net transmitters and Turkey, Vietnam and Argentina net receivers in the three systems estimated on returns, with a near-identical ranking (Spearman 0.96–0.99). These directions should not be read as evidence of lead–lag transmission. The Granger networks keep at most 2.5% of the possible lag links, so the data contain very little directional information; with negligible dynamics, the variance shares are close to symmetric squared correlations, and net connectedness mainly reflects the row normalization, that is, which markets are more or less correlated with the rest of the system.

A market such as Turkey or Vietnam, weakly correlated with the others, mechanically appears as a receiver. Trading hours add a second source of bias, since all funds trade in New York after the Asian markets have closed. The idiosyncratic network gives a different ranking (Spearman 0.30 with Diebold–Yilmaz): Taiwan and Indonesia are the main transmitters and Argentina is close to neutral. Argentina is thus exposed to the common emerging-market shock rather than to shocks from specific neighbors, and its largest country-specific shock, the August 2019 primary elections, does not turn it into a relevant transmitter in any of the four systems: in our data, that local shock did not spread.

Limitations. First, we study returns, not the realized volatilities of the base paper, so our networks describe the transmission of return shocks, not of risk. Second, ETF prices embed exchange-rate movements against the dollar and US trading hours, which affect the estimated directions. Third, the rolling windows of 104 weekly observations are short for the sparse estimators, whose results depend on tuning choices: NETS is selected by BIC over a grid of 15 values, which makes the series jump from one window to the next, and FNETS requires a positive-definite adjustment of the idiosyncratic covariance, gives an empty network in the full sample, and shows one-window drops that coincide with changes in the number of retained coefficients. With $N = 18$, the cross-section is also small for the factor-number criterion and for the consistency of the dynamic principal components, which rely on large N. Fourth, almost no lagged links survive in the sparse methods, so the Granger networks carry little information. Fifth, consecutive windows share 100 of their 104 weeks, so the stability measures are high partly by construction; they are informative for comparing methods but not as absolute values. Finally, we report no confidence intervals, so small differences, such as 81.0% against 80.0% between Diebold–Yilmaz and the lasso VAR, should not be interpreted as significant.

7 Conclusion

Measured connectedness among emerging equity markets depends less on the estimation method than on whether the common factor is removed. At the weekly frequency connectedness is high, around 80% for the dense measures, and almost entirely contemporaneous; the lasso VAR and Diebold–Yilmaz agree closely, and NETS reproduces the same pattern at a lower level. A single common factor explains more than half of the variance of returns, and removing it lowers connectedness by about 40%, a share that is stable across windows, horizons and samples; the peak of 2020 and the decline afterwards reflect changes in the weight of this factor rather than in bilateral links. Most of the co-movement of emerging markets is therefore common exposure, not contagion. What cannot be claimed with the same confidence is the direction of spillovers between countries, which our data barely identify, and the structure of the idiosyncratic network, which is unstable in short windows. Natural extensions are to study volatility connectedness, as in the base paper, to use higher-frequency data or longer windows for the factor-adjusted estimators, and to add more markets, which would also improve the estimation of the common factor.

Statement on the use of AI tools

The authors used AI tools in the following ways during the preparation of this paper. Claude Deep Research (Anthropic) was used to search for and identify sources and data. Claude Code (Anthropic) was used to write the code that produced the estimations, figures and tables, to typeset the paper in LaTeX, and to draft parts of the text under the authors' direction. The research questions, the structure of the paper, the main modeling decisions, the Discussion and the Conclusion are the work of the authors. The authors reviewed and revised all drafted text, agree with its content, and take full responsibility for the paper.

8 References

Matteo Barigozzi and Christian Brownlees. NETS: Network estimation for time series. *Journal of Applied Econometrics*, 34(3):347–364, 2019.

Matteo Barigozzi, Haeran Cho, and Dom Owens. FNETS: Factor-adjusted network estimation and forecasting for high-dimensional time series. *Journal of Business & Economic Statistics*, 42(3):890–902, 2024.

T. Tony Cai, Weidong Liu, and Harrison H. Zhou. Estimating sparse precision matrix: Optimal rates of convergence and adaptive estimation. *The Annals of Statistics*, 44(2):455–488, 2016.

Jiahua Chen and Zehua Chen. Extended Bayesian information criteria for model selection with large model spaces. *Biometrika*, 95(3):759–771, 2008.

Francis X. Diebold and Kamil Yılmaz. On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134, 2014.

Kristin J. Forbes and Roberto Rigobon. No contagion, only interdependence: Measuring stock market comovements. *The Journal of Finance*, 57(5):2223–2261, 2002.

Mario Forni, Marc Hallin, Marco Lippi, and Lucrezia Reichlin. The generalized dynamic-factor model: Identification and estimation. *Review of Economics and Statistics*, 82(4):540–554, 2000.

Marc Hallin and Roman Liška. Determining the number of factors in the general dynamic factor model. *Journal of the American Statistical Association*, 102(478):603–617, 2007.

Gary Koop, M. Hashem Pesaran, and Simon M. Potter. Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1):119–147, 1996.

Dom Owens, Haeran Cho, and Matteo Barigozzi. fnets: An R package for network estimation and forecasting via factor-adjusted VAR modelling, 2024. R package version 0.1.6, CRAN.

Jie Peng, Pei Wang, Nengfeng Zhou, and Ji Zhu. Partial correlation estimation by joint sparse regression models. *Journal of the American Statistical Association*, 104(486):735–746, 2009.

H. Hashem Pesaran and Yongcheol Shin. Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1):17–29, 1998.

Hélène Rey. Dilemma not trilemma: The global financial cycle and monetary policy independence. In *Proceedings of the Economic Policy Symposium, Jackson Hole*, pages 285–333. Federal Reserve Bank of Kansas City, 2013.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288, 1996.

Hui Zou. The adaptive lasso and its oracle properties. *Journal of the American Statistical Association*, 101(476):1418–1429, 2006.